Name: Gentle Steps -

A Camera free home activity monitor for elderly care.

Members (maximum 8 members):

S. No.	Name	Role / Contribution	Contact (Optional)
1.	Girish Suku Kumar	Architect and Implementation	
2.			
3.			
4.			

Project Checklist

Phase 1

- Project Title
- Cover Image
- List of Components
- Detailed Project Description
- Schematics
- Code
- CAD/ PCB Design If any *Optional
- Photo of project
- Video displaying working of Project
- BONUS/OPTIONAL: Video explaining working of the project

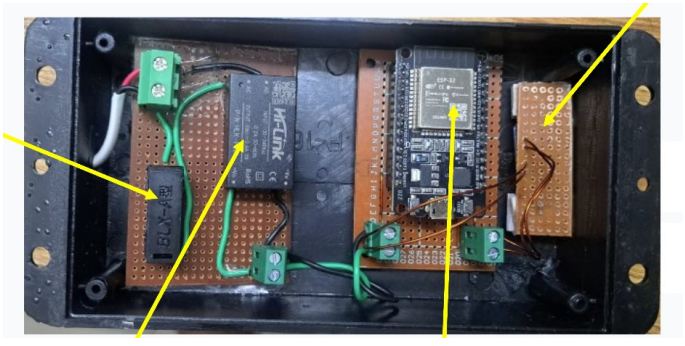
Gentle Steps -

1. A Camera free home activity monitor for elderly care.

Cover Image:







2 Component List:

ID	Name	Designator	Quantity
1	ESP32 Basic Dev Module	ESP32_Basic	4
4	LD2410c Presence Sensor	LD2410c	4
5	Hi Link 240V AC to 5V DC convertor	Power_supply	4
6	Vacuum Fuse	Vacuum Fuse	4
7	ESP32 S3 M5 Stack	M5_Stack	1
8	USB Power supply for ESP32S3-M5Stak	Power_supply_2	1
9	Raspberry Pi	Rpi_1	1
10	AWS Light Sail Instance	AWS	1

3. Project Explanation Step By Step :

This project implements an end to end system, that is used for monitoring Elderly people who are living independently in villas, apartments, Senior Citizen homes etc. It can used in hospitals too to observe the movement of patients, irrespective of age. End users will be care takers and close relatives (Eg kids) of the senior citizens

The project aimed to find answers the following questions.
1. How active are my parents who are living remotely.
2. Where are they in the house at presents
3. What are the movement patterns within the house
4. How often they are waking up a night to use the bathroom.
5. How often they are using bathroom during day.
6. Are there any indications of dehydration
7. What is the typical duration they are spending the bathroom
8. How are they sleeping, are they restless during sleep or getting a sound sleep ?
9. How long are they sleeping during day and night.
10. How often they go out of house.

All these questions can be answered by, installing a simple presence sensor (LD2410c) connected with ESP-32 inside the room. This sensor collects a whole lot of information about the human presence inside room LD 2410c can collect the following information about the human presence
1. Human presence YES/NO
2. Is the Human stationary of moving
3. At what distance the human presence was detected from the sensor.

When these three piece of information (coming from each room) is combined with time stamp, we can find answers to all the above questions and beyond. This is the exactly the objective of this project.

The data collected from each room is aggregated at a **Central unit** and processed with the help of Machine Learning algorithms, it provides a lot of insights into dynamics of the people living in the house.

Following is the overall Architecture or design of the system:

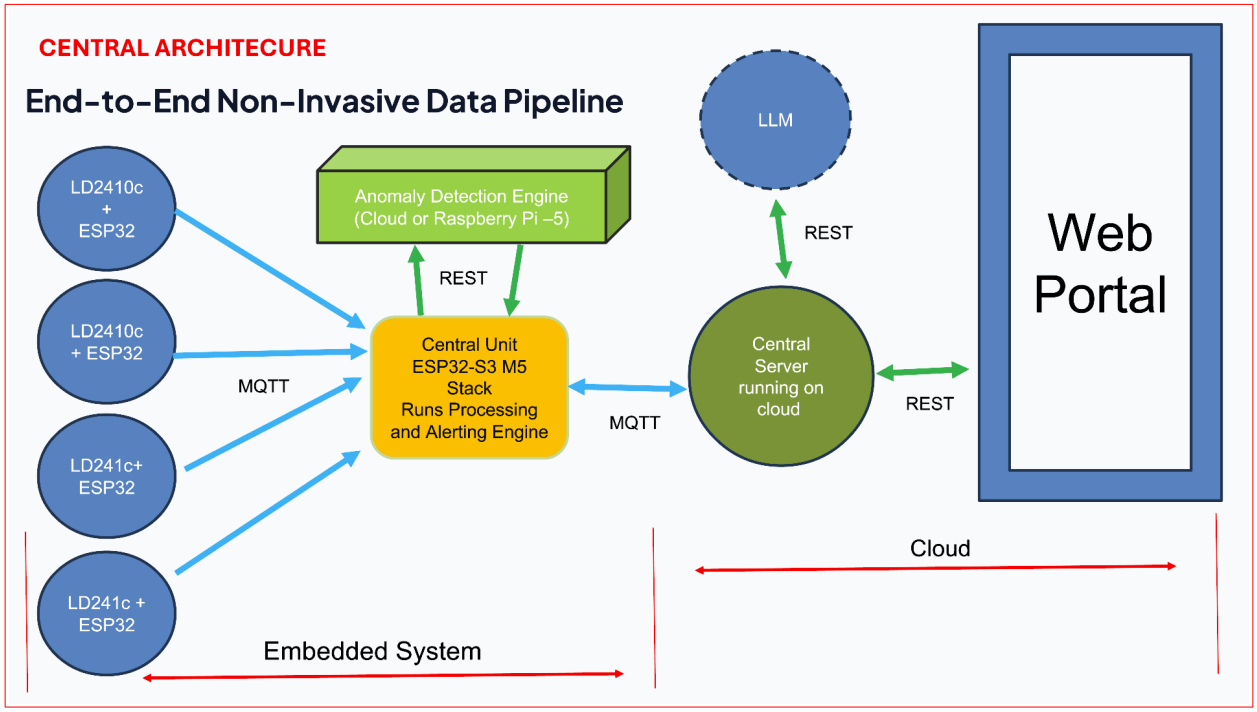


Fig-1 Overall Architecture of the system (REST → REST API, MQTT → Message queue Telemetry Transport)

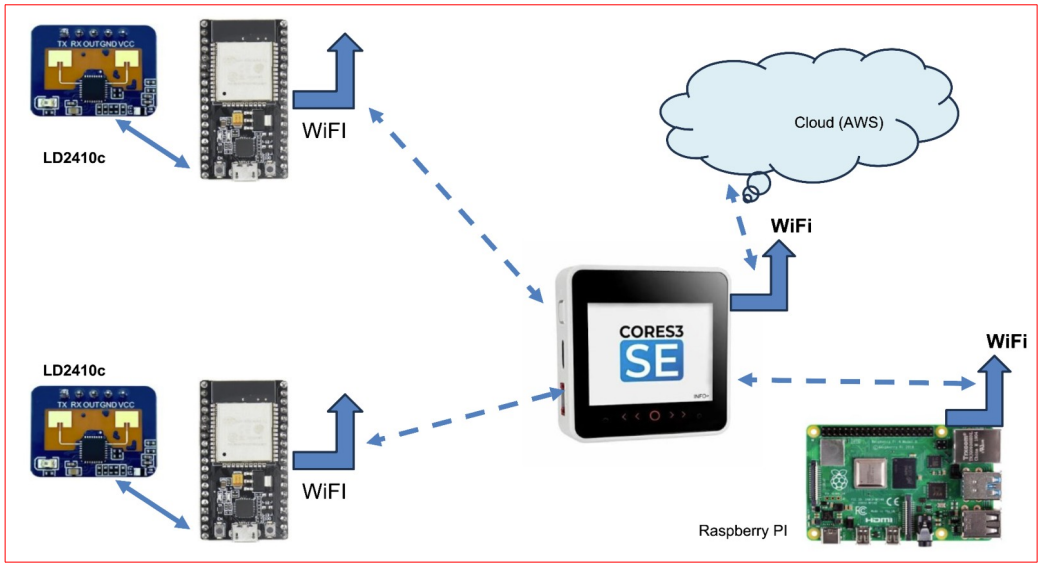
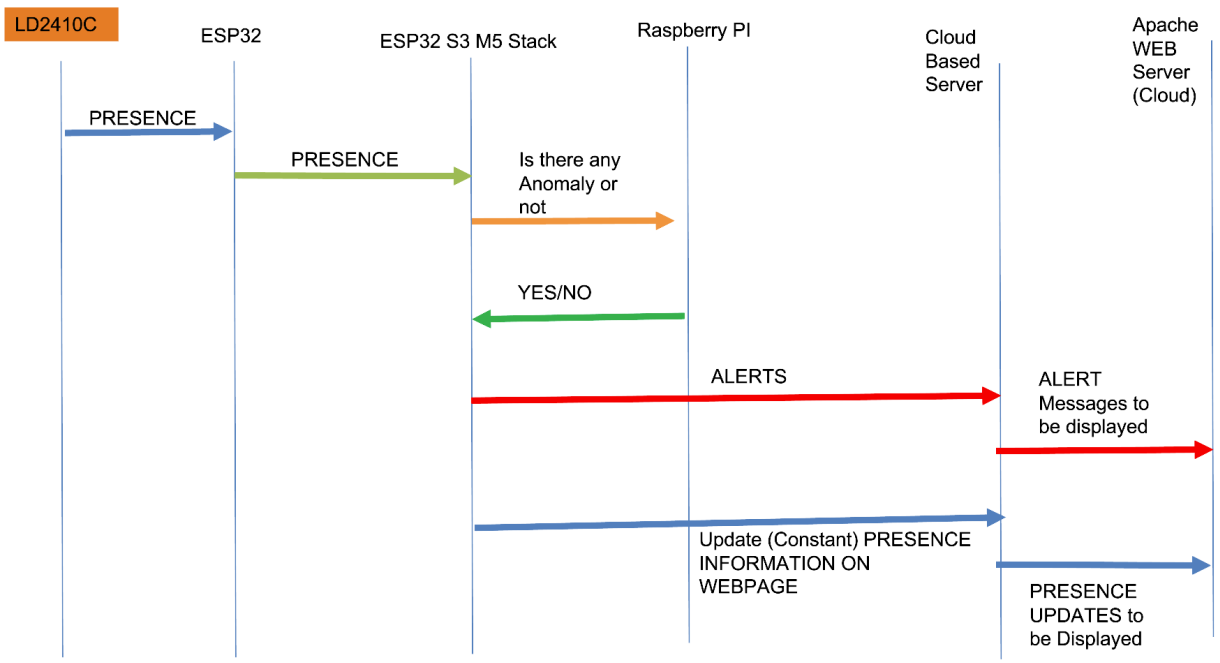


Fig-2 How various Hardware Components are connected over wireless using WiFi

Flow of Information between various Components (Message Sequence Diagram)



4.0 Coding Explanation and Configuration

First and foremost, we need to install the necessary library to the Arduino IDE. To do this, navigate to the Sketch menu and click on “Manage Library.” Next, search for the library name and click on the “Install” button.

There are two independent sketches

1. The sketch for the basic ESP32 unit which collects data from LD2410c and send it to the Central Unit for processing and Alert Generation running on ESP32 S3 M5 Stack

2. The Central Unit which collects data from various Sensors (ESP32 + LD2410C) and processes it and update Web Server

The following are the list of libraries that need to be installed for compiling the sketch for Central Unit that runs on ESP32 S3 M5 Stack

Go to Tools > Manage Libraries... and install:

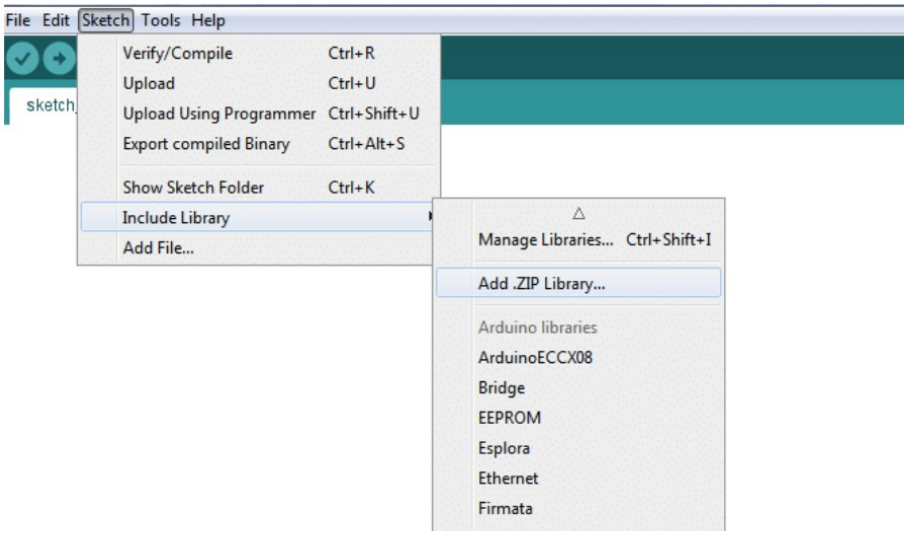
- `ArduinoJson`
- `WebSerial`
- `M5Unified`

- `ESP32MQTTClient`

The following are the list of libraries that need to be installed for compiling the sketch for Sensir Unit (**ESP32 + LD2410C)** Unit.

Go to Tools > Manage Libraries... and install:

- `ArduinoJson`
- `WebSerial`
- **MyLD2410** `(github)`
- `ESP32MQTTClient`



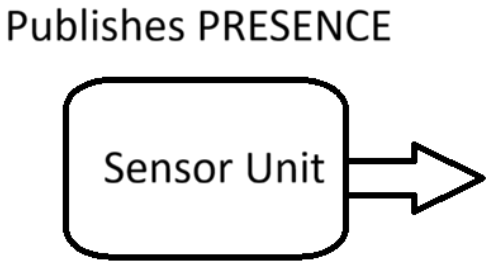
The two sketches has to be kept in to difference folders and each of them has to complied and flashed separately.

(I) The source code the Central Unit can you found in the GitHUB Repository
https://github.com/girishsukukumar/PRESENCE_CPU_WEB

(ii) The source code for the Sensor Unit can be found in the GITHUB Respository
https://github.com/girishsukukumar/LD2410c_SENSOR_UNIT

5.0 **Details of Sensor Unit (ESP 32 + LD2410c)**

Important part of the source code for the Sensor Unit




```
  while (true)
  {
     if (sensor.check() == MyLD2410::Response::DATA)
     {
        if (sensor.presenceDetected())
        {
           currentPresenceState =  PRESENSE_DETECTED;
          presence = "DETECTED"
        }
        else
        {
           currentPresenceState = NO_PRESENCE;
           presence = "NOT_DETECTED" ;
        }

        if (lastStateOfPresence != currentPresenceState)
        {
            statusChanged = true;
            lastStateOfPresence = currentPresenceState;

        }

  // Reflect current reading each pass instead of latching forever
  StationaryTargetDetected = sensor.stationaryTargetDetected();
  StationaryTargetDistance = StationaryTargetDetected ?
sensor.stationaryTargetDistance() : -1;

  MovingTargetDetected = sensor.movingTargetDetected();
  MovingTargetDistance = MovingTargetDetected ? sensor.movingTargetDistance()
: -1;

  // Publish ONLY on a genuine state transition (NO_PRESENCE <->
PRESENCE_DETECTED)

  bool shouldPublish = mqttClient.isConnected() && statusChanged;
     if (shouldPublish)

      {
```

```
        long rssi = WiFi.RSSI();

        lastIdleMessage = millis();

        int written = snprintf(mqttMessage, sizeof(mqttMessage),

          "{\"sensor_room\":\"%s\",\"status\":\"%s\","

          "\"Moving Target\":%s,\"Moving Target Dist\":%d,"

          "\"Stationary Target\":%s,\"Stationary Target Dist\":%d, \"Wifi\":
%ld}",

          ConfigData.wifiDeviceName,

          presence.c_str(),

          MovingTargetDetected     ? "true" : "false",

          MovingTargetDistance,

          StationaryTargetDetected ? "true" : "false",

          StationaryTargetDistance,

          rssi

        );
```

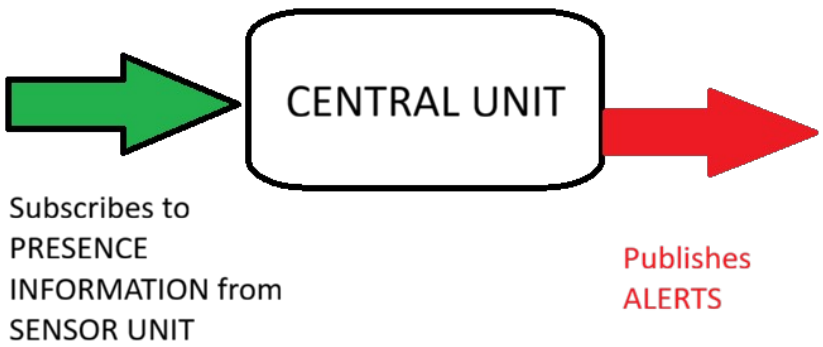
6.0 Details of Central Unit (ESP32 S3 M5 Stack)

Following are the Details of code running on Central Unit (ESP32 S3 M5 Stack**)**

Central Unit has two jobs

1. Central Unit acts as MQTT Subscriber, subscribing to message from Sensor Unit

2. Acts as publisher Publishing ALERTS.



```
void onDataReceived(const std::string &topic, const std::string &payload)
{
char message[200]; JsonDocument doc; bool detected ;
DeserializationError error = deserializeJson(doc, payload); if
(xSemaphoreTake(bathroom_mutex, pdMS_TO_TICKS(500)) == pdTRUE) { if
(error) { Serial.print("JSON parse failed:
"); Serial.println(error.c_str()); return; } noOfMQTTMessages++ ;


 const char* room   = doc["sensor_room"];
 const char* status = doc["status"];
 bool movingTarget     = doc["Moving Target"];
 bool stationaryTarget = doc["Stationary Target"];
 char event[10] ; // To be incorportated later


 int movingDist     = doc["Moving Target Dist"];
 int stationaryDist = doc["Stationary Target Dist"];

 int idx = findRoomIndex(room);

 if (idx < 0)
 {
 DEBUG_PRINTF(" Wrong room name received %s \n",room ) ;
 return;
 }

 time_t now = getEpochTime();
 roomState[idx].lastMsgEpoch = now;
 if ((strcmp(status, "DETECTED") == 0) && (roomState[idx].occupied == false))
  {
    //Serial.printf("%s :status == DETECTED and occupied == False\n",room) ;
      roomState[idx].occupied = true;
      roomState[idx].occupiedSinceEpoch = now;
      DEBUG_PRINTF("%s , roomState[idx].occupiedSinceEpoch = %d\n ",room,
now) ;

        if (strcmp(room, "Toilet-1")==0)
        {
              DEBUG_PRINTF("Presence detected in Toilet-1\n");
          SetNewDataArrivedFlag() ;

              if (IsDayOrNight(getEpochTime()) == true)
              {
             IncrementBathRoomDayVisits() ;
             PrintEpochToHumanReadableFormat(now) ;
             RecordtimeOfBathroomDayVisit(now) ;
             PrintEpochToHumanReadableFormat(GetTimeOfBathroomDayVisit());
```

```
            DEBUG_PRINTF("IncrementBathRoomDayVisits\n") ;
             }
             else
             {
            IncrementBathRoomNightVisits() ;
            PrintEpochToHumanReadableFormat(now) ;
            RecordtimeOfBathroomNightVisit(now) ;
            PrintEpochToHumanReadableFormat(GetTimeOfBathroomNightVisit());
            DEBUG_PRINTF("IncrementBathRoomNightVisits\n") ;
             }
         PrintBathRoomUsageData();

   }


     if (strcmp(room, "BedRoom")==0)
     {
              DEBUG_PRINTF("Presence detected in BedRoom\n");

            if (IsDayOrNight(getEpochTime()) == false)
            {
               DEBUG_PRINTF("Presence detected in BedRoom at night\n");
               /* Track movements only during night */
                  if (movingTarget == true)
                  {
                     incrementBedRoomMovementCounter() ;
                  }
             }
    }

}
else if ((strcmp(status, "NOT_DETECTED") == 0) &&
         (roomState[idx].occupied == true))
{
    //Serial.printf("%s: status == NOT_DETECTED and occupied == true\n",room)
;
    roomState[idx].occupied = false;
    WhenRoomIsEmpty(room, roomState[idx].occupiedSinceEpoch, now);
}

else if ((strcmp(status, "KEEP_ALIVE") == 0))
{
    struct tm currentTimeDate ;
    int hour ;
    int minutes ;
    roomState[idx].lastMsgEpoch = time(nullptr);
    //Get current time,
    currentTimeDate = convertEpochToHumanReadableFormat(getEpochTime()) ;
    hour =  currentTimeDate.tm_hour ;
    minutes = currentTimeDate.tm_min ;

    if (hour == 11)
    {
      if (minutes > 55)
      {
          char message[100];
          /* Generate room wise break up report */
```

```
          int TimeSpendinLivingRoom = roomState[0].timeSpend;
          int TimeSpendinBedRoom    = roomState[1].timeSpend ;
          int TimeSpendinToilet     = roomState[2].timeSpend ;
          int TimeSpendinKitchen    = roomState[3].timeSpend ;
          int totalTime = TimeSpendinLivingRoom + TimeSpendinBedRoom +
TimeSpendinToilet + TimeSpendinKitchen ;

          float percentageSpendLivingRoom  =
(TimeSpendinLivingRoom/totalTime)/100 ;
          float percentageSpendBedRoom     =
(TimeSpendinBedRoom/totalTime)/100 ;
          float percentageSpendToilet      =
(TimeSpendinToilet/totalTime)/100 ;
          float percentageSpendKitchen     =
(TimeSpendinKitchen/totalTime)/100 ;

          snprintf(message,sizeof(message),
                   "%d-%d-%d , %f , %f , %f , %f", currentTimeDate.tm_mday,
                                                   currentTimeDate.tm_mon,
                                                   currentTimeDate.tm_year,
                                                   percentageSpendLivingRoom,
                                                   percentageSpendBedRoom,
                                                   percentageSpendToilet,
                                                   percentageSpendKitchen) ;

          mqttClient.publish(ConfigData.REPORT_TOPIC, message, 0, false);
          /* Clearing the collected data once in a day */
          /* After the statistics are geneared */

          InitBathRoomUsageData() ;
          InitBedRoomData() ;
          ReSetNewDataArrivedFlag() ;
      }
    }
}

xSemaphoreGive(bathroom_mutex);

}
else
{
   DEBUG_PRINTF("Unable to get mutex to update Room status\n") ;
}

}



void WhenRoomIsEmpty(const char* room, time_t startEpoch, time_t endEpoch)
{
  int timeSpendinRoom = (int)(endEpoch - startEpoch);

  int idx = findRoomIndex(room);
  roomState[idx].timeSpend =  roomState[idx].timeSpend + timeSpendinRoom ;
```

```
if (strcmp(room, "Toilet-1") == 0)
{ // Real-time per-visit duration check. // NOTE: the bathroom sensor on the
leaves is named "Toilet-1" (there is // no "BathRoom" room); this branch was
previously dead because the // string never matched.


if (IsDayOrNight(getEpochTime()) == true)
  {
   if (timeSpendinRoom < 2)
   {
      // if time spend in any room is less that 5 secs then
      // it is noise from a stray signal or reflection
      // Found this during testing
      DEBUG_PRINTF("Time Spend in Toilet-1 is %d\n",timeSpendinRoom);
      DecrementBathRoomDayVisits();
   }
   else
       {
     UpdateBathRoomDayVisitDuration(timeSpendinRoom);
   }
  }
  else
  {
   if (timeSpendinRoom < 5)
   {
      // if time spend in any room is less that 5 secs then
      // it is noise from a stray signal or reflection
      // Found this during testing
      DEBUG_PRINTF("Time Spend in Toilet-1 is %d\n",timeSpendinRoom);
      DecrementBathRoomNightVisits();
   }
   else
   {
         UpdateBathRoomNightVisitDuration(timeSpendinRoom);
   }
  }

PrintBathRoomUsageData();

}
if (strcmp(room, "BedRoom") == 0)
{




}
if (strcmp(room, "LivingRoom") == 0)

{

}

if (strcmp(room, "Kitchen") == 0)
{

}
```

```
}
```

There are two main functions in the Central Unit which keep tracks of entry and exit into each room

1. **onDataReceived**(), the MQTT callback which gets executed when each time a PRESENCE message arrives from the sensor unit.

2. **WhenRoomIsEmpty()** this function is called by **onDataReceived()** When a room becomes empty.

With the entry and exit timestamp,we can calculate the duration of time spend in each room and number of times the room was occupied during a day/night. This is the information which is used for checking whether there are any anomalies with the pattern on usage.

7.0 Anomaly Detection

The Central Unit run on ESP32 S3-M5 Stack. This unit does not have enough RAM to run a very basic anomaly detection based on Isolation forest Algorithm. For Isolation forest algorithm to be executed we need a machine which at least 512MB to 1 GB of RAM. This is exactly where a Raspberry Pi comes into picture. The anomaly detection program was developed in python. Central Unit Interacts with this Python code using REST APIs. The isolation forest implementation for anomaly detection is trained separately. The code which does anomaly detection and code which is used to create models for anomaly detection can be found in github at the following repository:
https://github.com/girishsukukumar/ISOLATION_FOREST

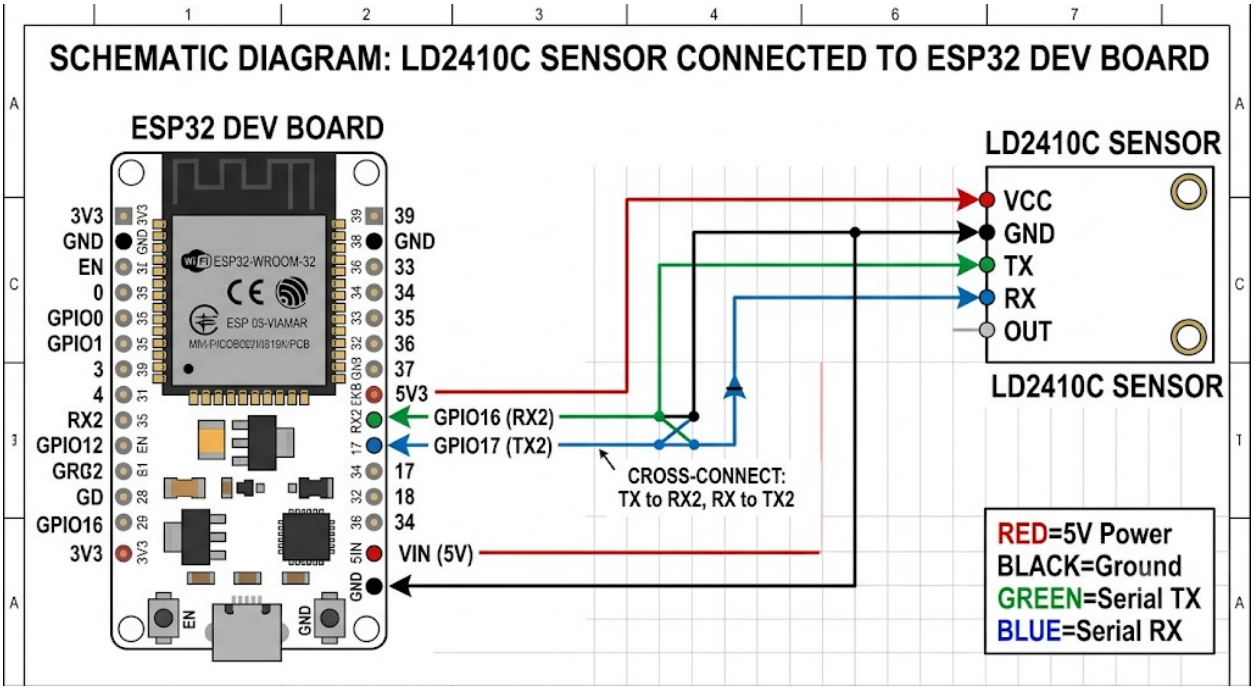
Since the anomaly detection module is written in python it is platform independent. It can be executed on any hardware which supports python. So this unit for anomaly detection can be executed on machine hosted on cloud too.

8.0 Front End Design

The front end of the web page is designed using React.JS. The graphical features of React.js is used to draw the layout display messages.

There are two web servers running on cloud. The first one is apache web server responsible for displaying the front end and other activities. Second webserver interacts with the Central Unit and receives message and send and send it to apache web server for displaying . The Second web server listen to port no 5000 and act as a MQTT Subscriber, which receives messages from Sensor Unit and Central Unit.

9.0 Schematics for the Sensor Unit

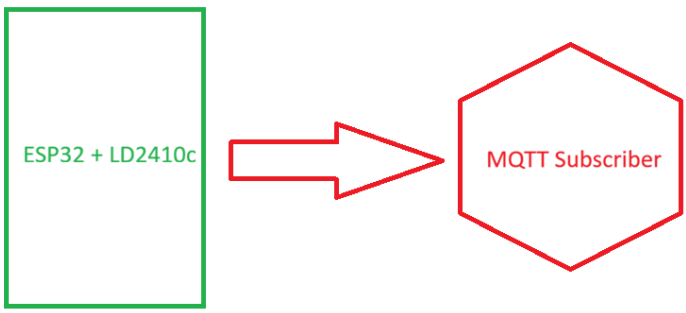


10.0 Testing

Testing of the whole unit is done in three steps.

1.Testing of LD2410c Sensor + ESP32 (Sensor Unit)

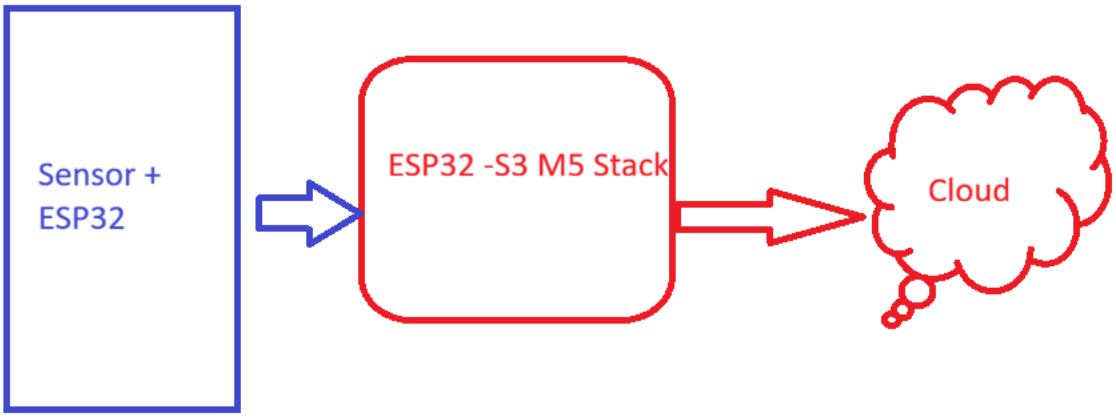
The Sensor unit was tested as a stand alone by keeping it in a room and observe human movements. Each time when there is change in PRESENCE the sensor unit send out a MQTT Message. This message was received by a MQTT subscriber Written in python. We verify the message manually. In addition to subscriber written in python, another Windows based application called MQTT Explorer was also used to receive and examine the messages.



2.Testing with real sensor installed across rooms

Objective of this activity is to integrate Sensor Unit with Central Unit and see how Central Unit is responding the data coming from sensor unit. This is a time consuming process as we need some one who can enter each room and exit after some time.

3.



System testing with simulator

Since testing with real sensor need resources and time a Simulator which simulate the PRESENCE information message from the Sensor Unit was developed in python. With this simulator the product was tested for long time to check the long time endurance of the system.



Source of the simulator developed in python can be found in github repository

https://github.com/girishsukukumar/PRESENCE_SYSTEM_TEST

11.0 PCB Files

There are no PCB files as the project was done using a general purpose PCB

12.0 Project Video:

https://tinyurl.com/gentlestepsefy

13.0 Project Presentation slides

girishsukukumar/AIOT_DEMO

14.Code:

(All the following respositories are public)

(I) The source code the Central Unit can you found in the GitHUB Repository
https://github.com/girishsukukumar/PRESENCE_CPU_WEB

(ii) The source code for the Sensor Unit can be found in the GITHUB Respository
https://github.com/girishsukukumar/LD2410c_SENSOR_UNIT

(iii) Source code for Anomaly detection
https://github.com/girishsukukumar/ISOLATION_FOREST

(iv) Source code for System Test code

https://github.com/girishsukukumar/PRESENCE_SYSTEM_TEST

15.0 Login Credentials for EFY and Digikey to see the portal live

Live Portal URL : **http://gentlesteps.i2otlabs.com/** **(Note : It is http:// and not https://)**

Login Name : teamefy

Password : digikey1410